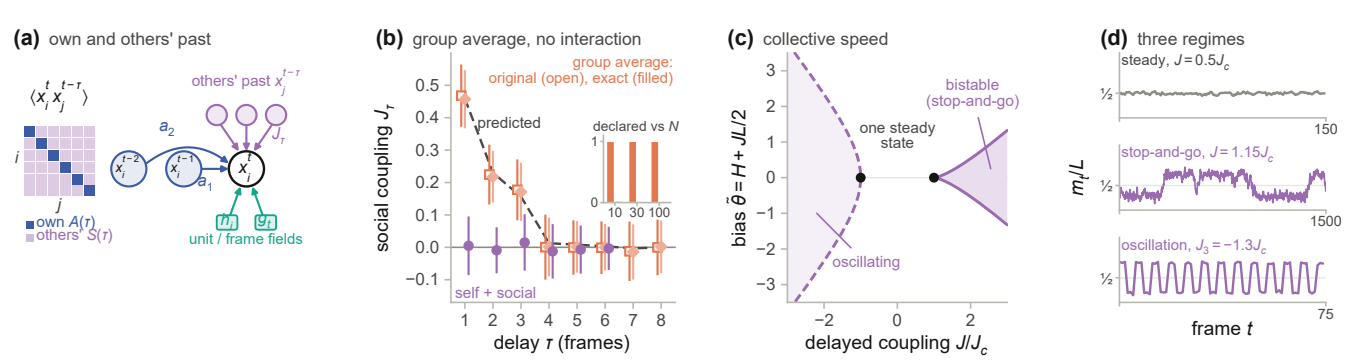

(a) own and others' past
⟨ x_i^t x_j^{t−τ} ⟩
i
j
own A(τ)
others' S(τ)
others' past x_j^{t−τ}
a_2
a_1
J_τ
x_i^{t−2}
x_i^{t−1}
x_i^t
h_i
g_i
unit / frame fields
(b) group average, no interaction
group average:
original (open), exact (filled)
predicted
declared vs N
1
0
10 30 100
social coupling J_τ
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
self + social
1 2 3 4 5 6 7 8
delay τ (frames)
(c) collective speed
bistable
(stop-and-go)
one steady
state
oscillating
bias θ̃ = H + JL/2
3
2
1
0
−1
−2
−3
−2 0 2
delayed coupling J/J_c
(d) three regimes
steady, J = 0.5J_c
½
150
stop-and-go, J = 1.15J_c
½
1500
oscillation, J_3 = −1.3J_c
½
75
m_t/L
frame t